

THROUGH THE AGES

Science

For centuries, brilliant women have worked tirelessly to discover and analyze the unknown processes of the universe. Nonetheless, women in science and mathematics have had to fight for their work to be recognized by the academic community.

▼ Flair for numbers

Émilie du Châtelet was a French scientist in the 18th century. She applied her talent at math to win at gambling, using the proceeds to buy books and laboratory equipment. She was Voltaire's mistress, and together they made Newton's work accessible for the rest of France.



SINCE ANTIQUITY, trailblazing women have pursued the studies of science and mathematics. Hypatia of Alexandria was the first documented female mathematician in 4th-century ancient Greece, where she taught philosophy and astronomy as head of the Neoplatonic School. Women worked as healers in ancient Egypt and, across diverse cultures, women have a long history of experimenting with herbs and other natural remedies. They practiced as midwives and nurses, and passed down their skills and knowledge from one generation to the next. In the Middle Ages, convents in Europe provided education for women—a crucial resource, as most of the world's earliest universities did not accept women students.

As the centuries passed, upper-class women pursued science and mathematics as a hobby. Some assisted their husband's research, while others struck out on their own. However, these pioneers were usually denied formal qualifications, and their work was often ignored. There were notable exceptions: in 1732 the Italian physicist Laura Bassi became the first woman to be admitted to the Bologna Academy of Sciences. She was awarded a doctorate and a chair (the first physics professor at a European university) but

“Nothing worthwhile comes **easily** ... In my day I was told **women** didn't go into **chemistry**.”

GERTRUDE ELION, NOBEL PRIZE WINNER, 1988

due to her gender she was not allowed to teach at the university, and had to give lectures at her home. Maria Agnesi was appointed as the first woman mathematics professor, also at Bologna, in 1750, but it is doubtful that she actually took up the post.

In the 19th century, more women began to work as scientists (see pp. 166–167). Mary Anning made discoveries in Jurassic marine fossil beds on the south coast of England that changed our understanding of prehistory and the Earth's history. Astronomers such as Caroline Herschel and Maria Mitchell used telescopes to see beyond our own planet and reveal the mysteries of the galaxy. Yet, as late as 1882, even eminent scientist Charles Darwin wrote: “I certainly think that women, though generally superior to men [in] moral qualities, are inferior intellectually.” Prestigious universities such as Oxford and